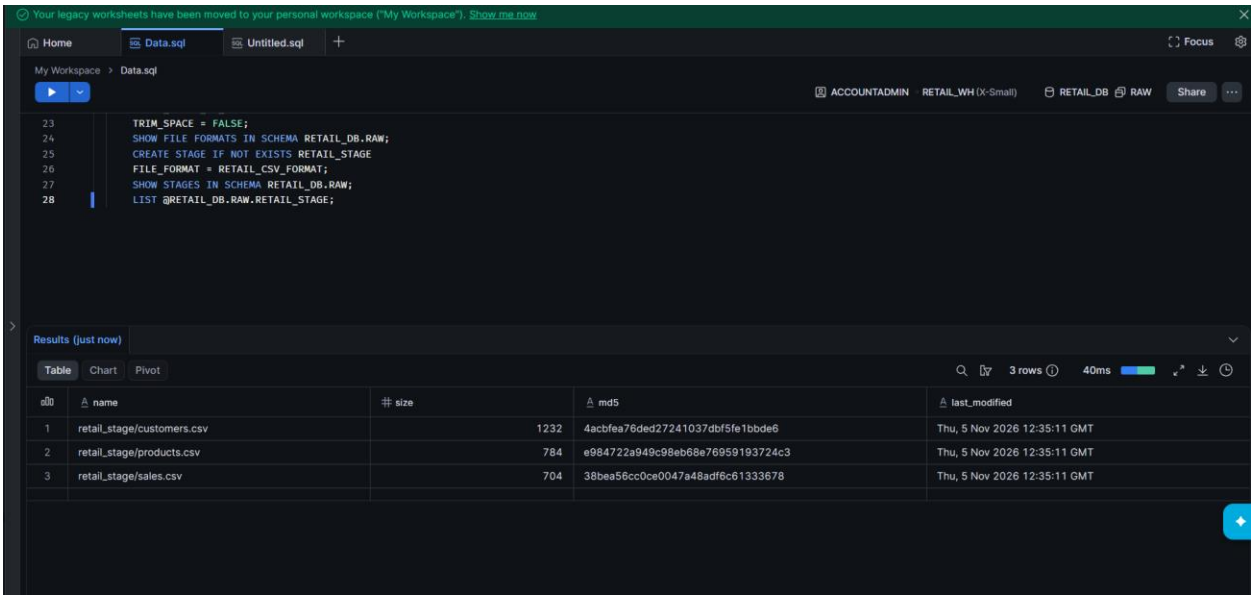
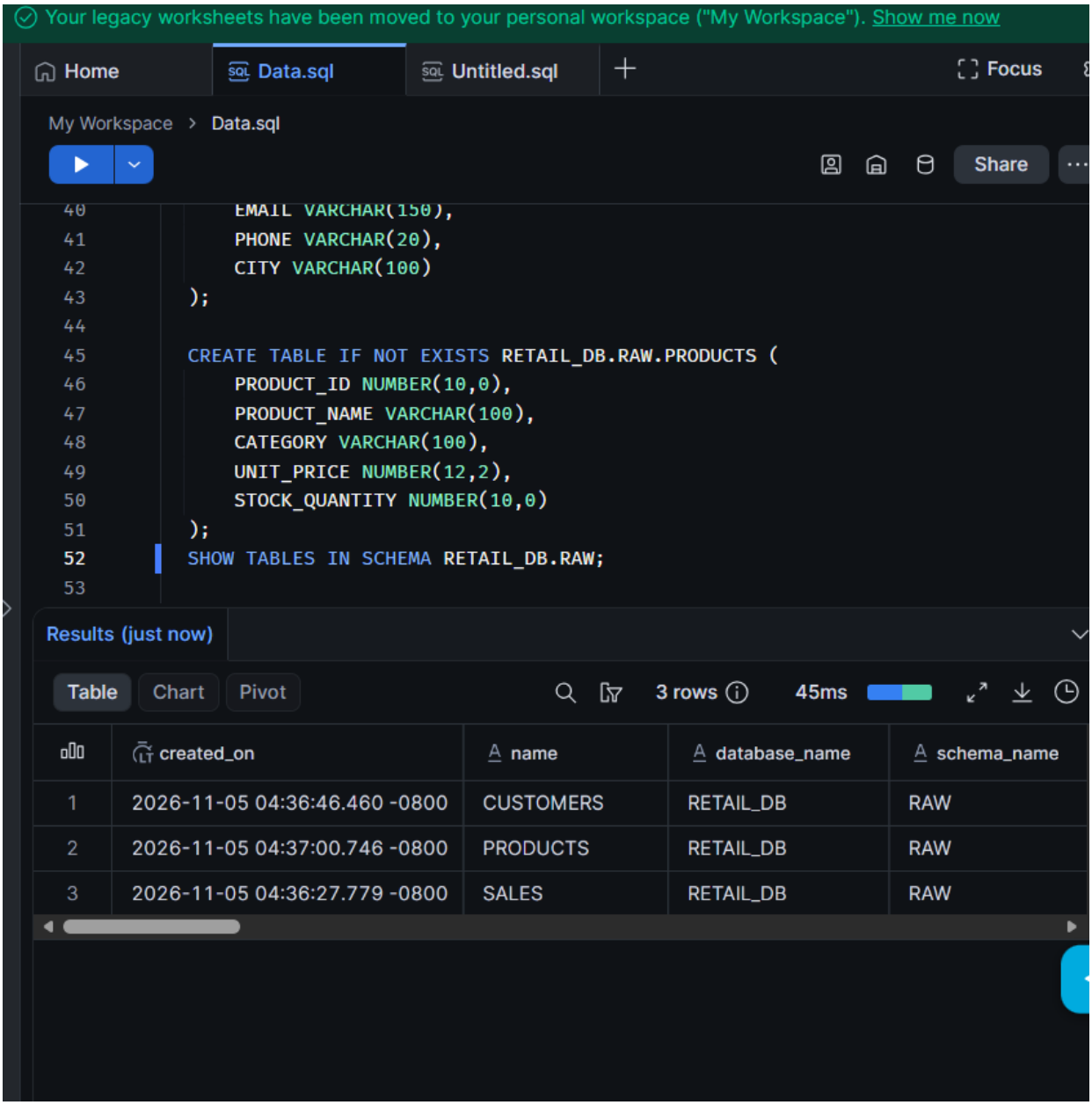
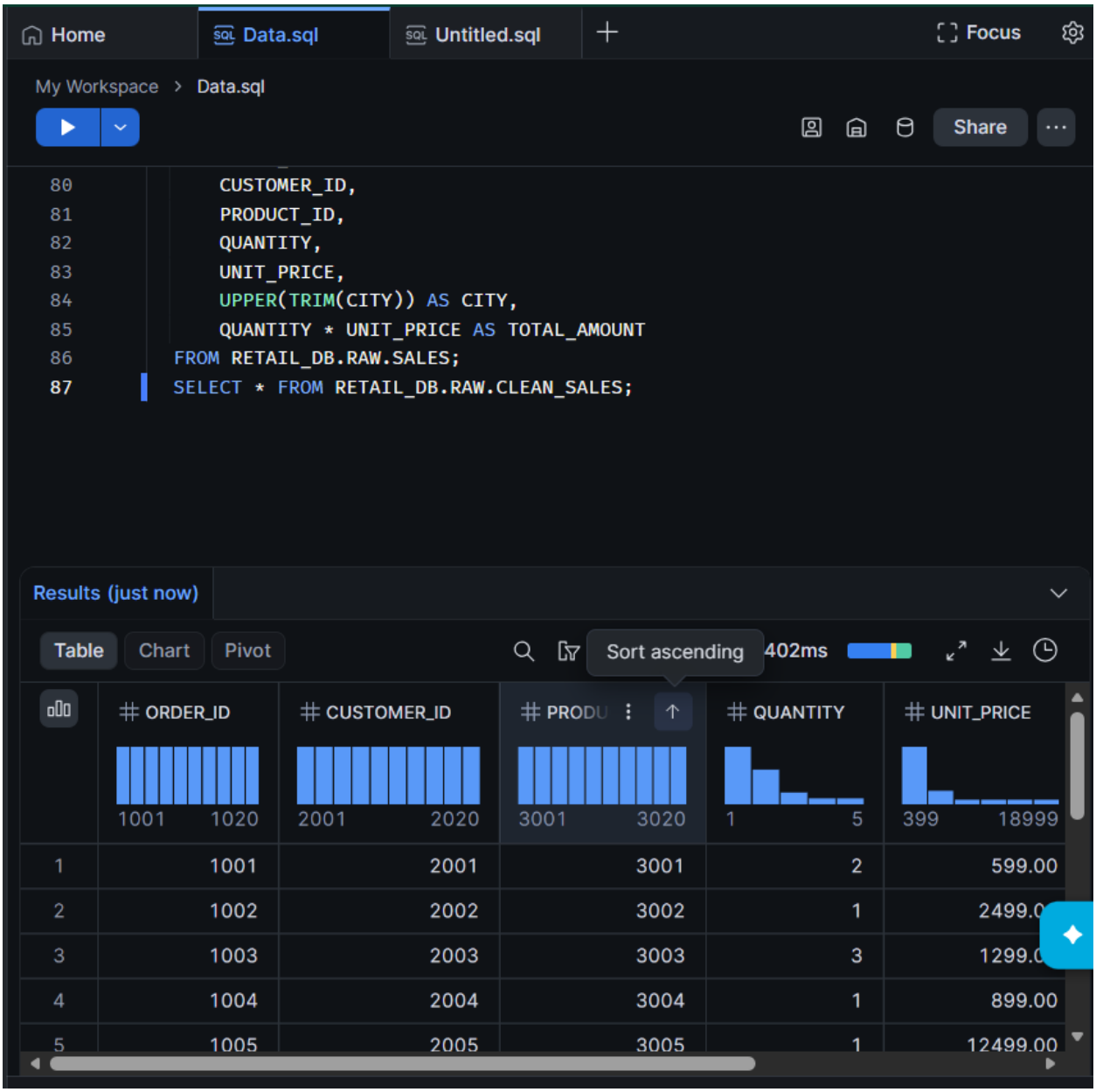
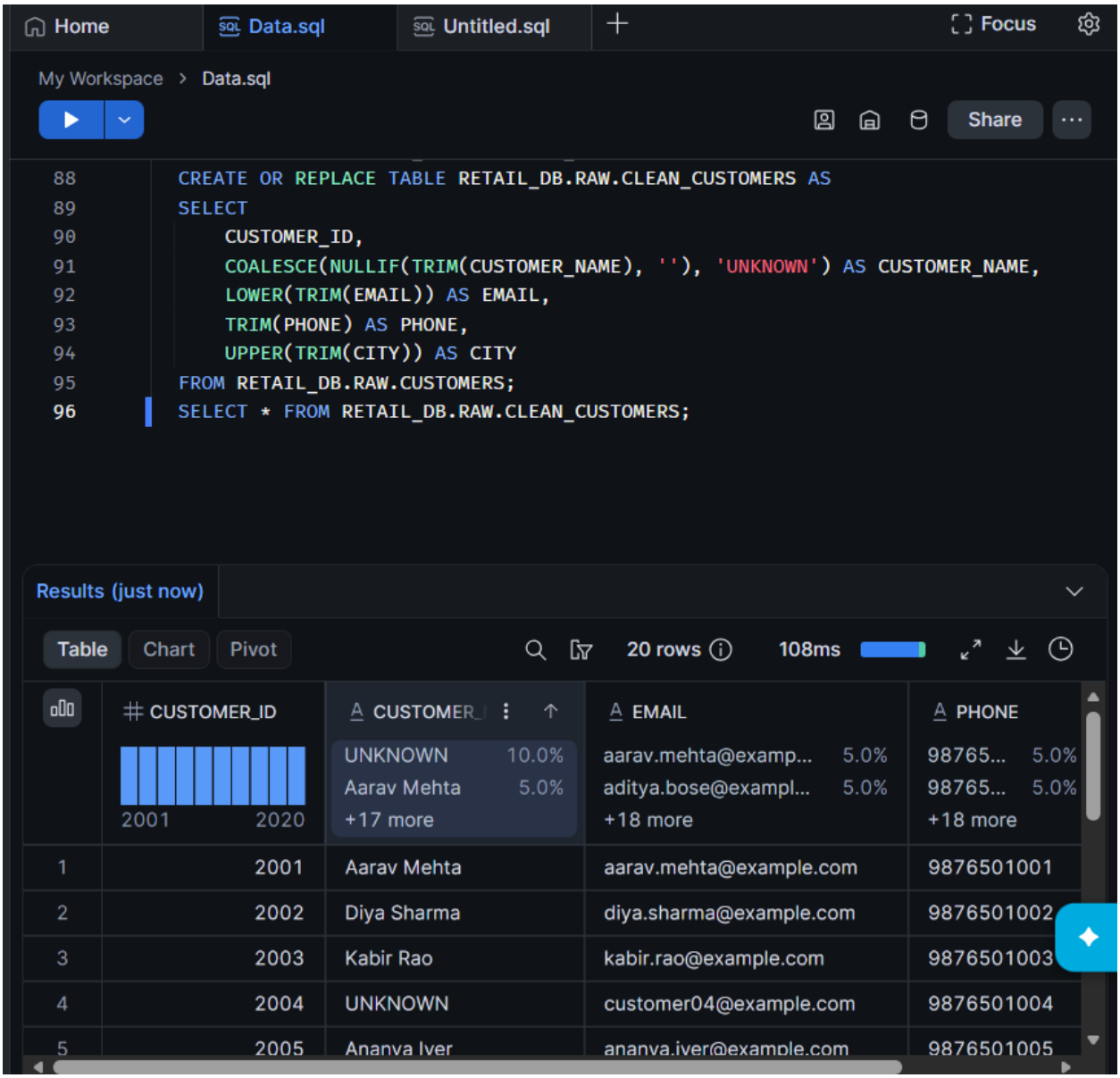
Snowflake – Multi-Dataset Ingestion, Transformation & Query Performance Optimization

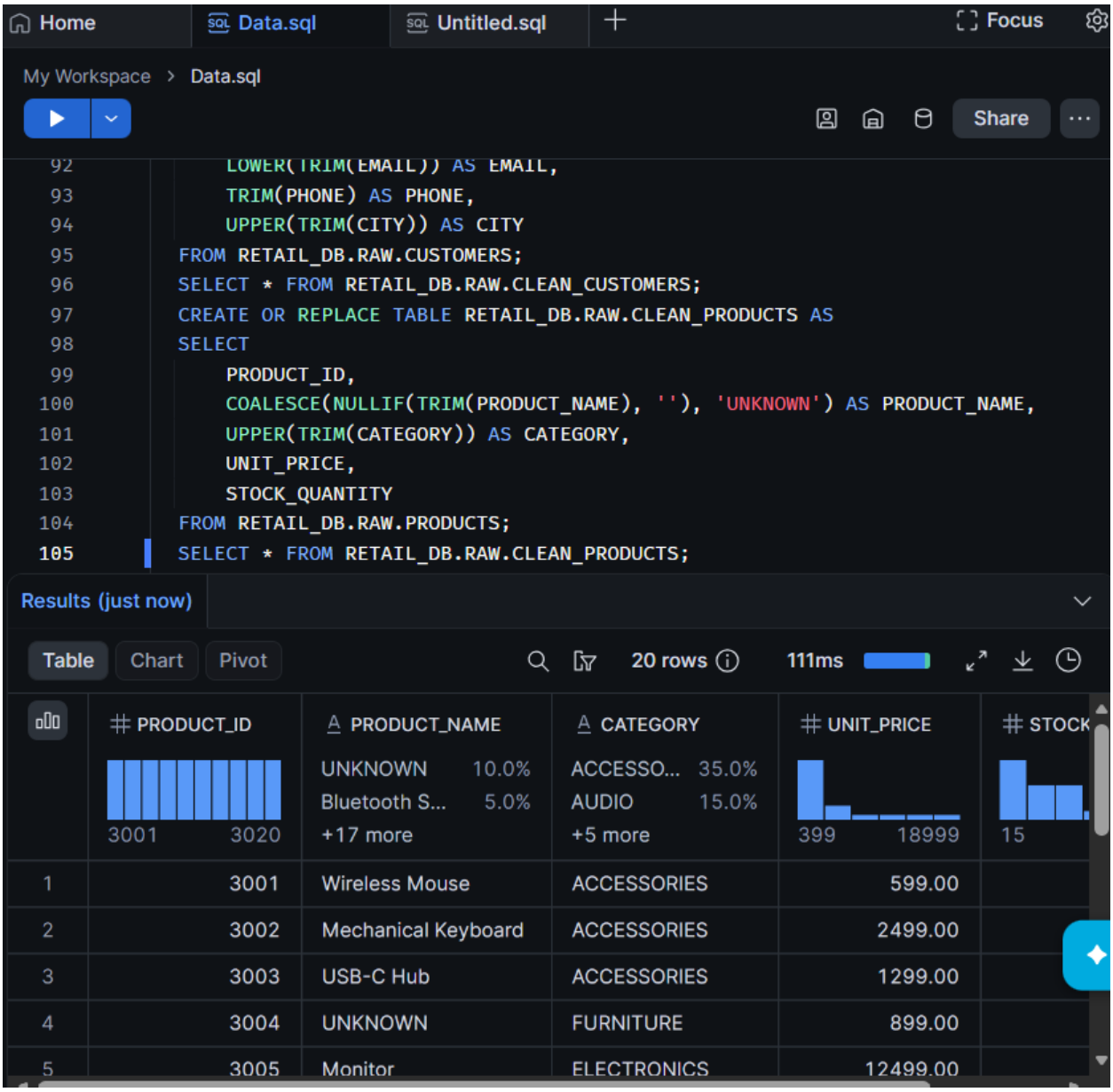
1. Prepare and ingest multiple related CSV datasets.
2. Create appropriate database tables with suitable column names and data types.
3. Configure file formats and internal stages for CSV ingestion.
4. Load datasets using Snowflake's COPY INTO command.
5. Validate data quality, record counts, duplicates, NULL values, and data types.
6. Clean and transform raw datasets using SQL.
7. Perform analytical queries involving joins, aggregations, grouping, and filtering.
8. Identify opportunities to optimize SQL query performance.

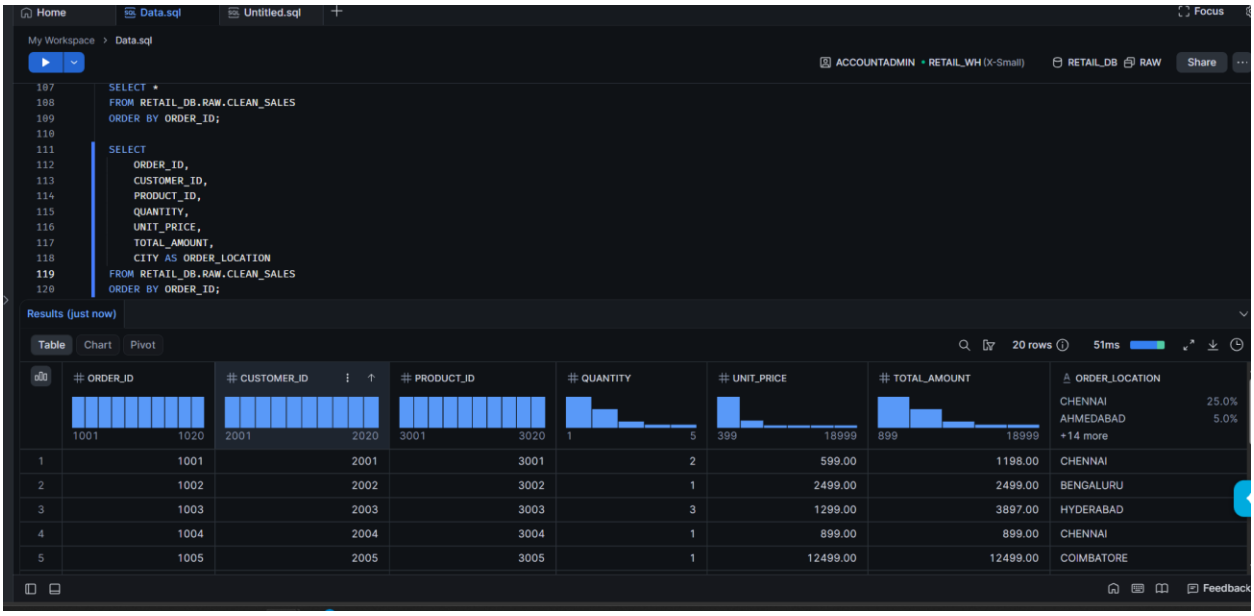

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Home
Data.sql
Untitled.sql
Focus
My Workspace > Data.sql
ACCOUNTADMIN
RETAIL_WH (X-Small)
RETAIL_DB
RAW
Share
TRIM_SPACE = FALSE;
SHOW FILE FORMATS IN SCHEMA RETAIL_DB.RAW;
CREATE STAGE IF NOT EXISTS RETAIL_STAGE
FILE_FORMAT = RETAIL_CSV_FORMAT;
SHOW STAGES IN SCHEMA RETAIL_DB.RAW;
LIST @RETAIL_DB.RAW.RETAIL_STAGE;
Results (just now)
Table
Chart
Pivot
3 rows
40ms
name
size
md5
last_modified
1 retail_stage/customers.csv 1232 4acbfea76ded27241037dbf5fe1bbde6 Thu, 5 Nov 2026 12:35:11 GMT
2 retail_stage/products.csv 784 e984722a949c98eb68e76959193724c3 Thu, 5 Nov 2026 12:35:11 GMT
3 retail_stage/sales.csv 704 38bea56cc0ce0047a48adf6c61333678 Thu, 5 Nov 2026 12:35:11 GMT

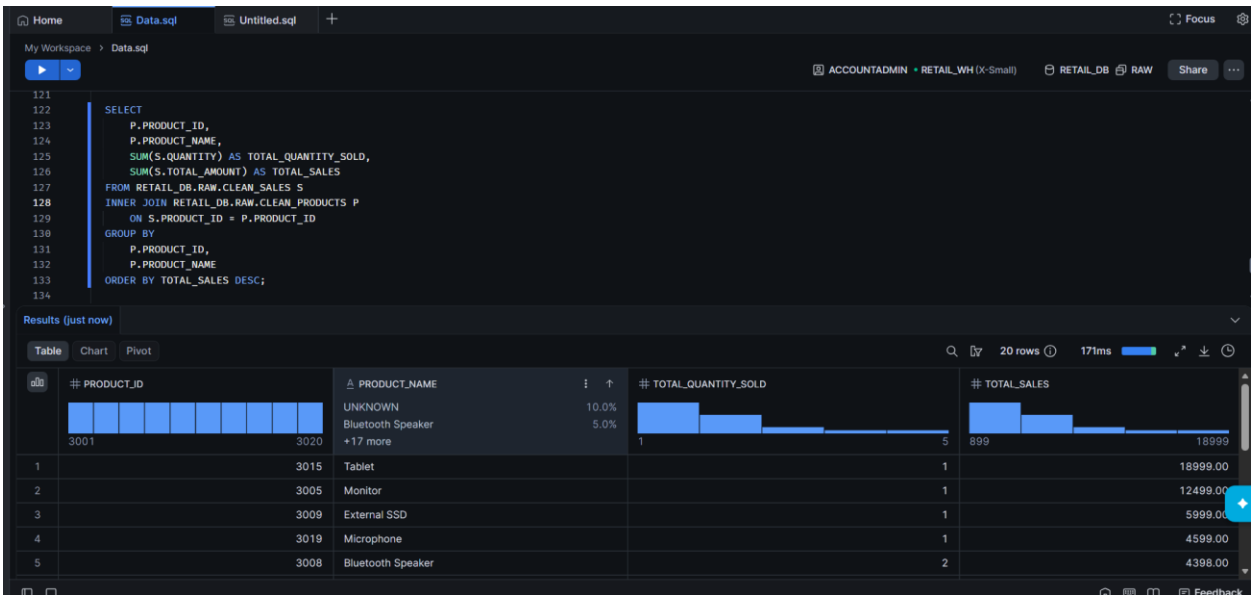

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Data.sql
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Focus
My Workspace > Data.sql
Share
EMAIL VARCHAR(150),
PHONE VARCHAR(20),
CITY VARCHAR(100)
);
CREATE TABLE IF NOT EXISTS RETAIL_DB.RAW.PRODUCTS (
PRODUCT_ID NUMBER(10,0),
PRODUCT_NAME VARCHAR(100),
CATEGORY VARCHAR(100),
UNIT_PRICE NUMBER(12,2),
STOCK_QUANTITY NUMBER(10,0)
);
SHOW TABLES IN SCHEMA RETAIL_DB.RAW;
Results (just now)
Table
Chart
Pivot
3 rows
45ms
created_on
name
database_name
schema_name
1 2026-11-05 04:36:46.460 -0800 CUSTOMERS RETAIL_DB RAW
2 2026-11-05 04:37:00.746 -0800 PRODUCTS RETAIL_DB RAW
3 2026-11-05 04:36:27.779 -0800 SALES RETAIL_DB RAW


Home
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My Workspace
Data.sql
Share
80 CUSTOMER_ID,
81 PRODUCT_ID,
82 QUANTITY,
83 UNIT_PRICE,
84 UPPER(TRIM(CITY)) AS CITY,
85 QUANTITY * UNIT_PRICE AS TOTAL_AMOUNT
86 FROM RETAIL_DB.RAW.SALES;
87 SELECT * FROM RETAIL_DB.RAW.CLEAN_SALES;
Results (just now)
Table
Chart
Pivot
Sort ascending
402ms
ORDER_ID
1001
1020
CUSTOMER_ID
2001
2020
PRODU
3001
3020
QUANTITY
1
5
UNIT_PRICE
399
18999
1 1001 2001 3001 2 599.00
2 1002 2002 3002 1 2499.0
3 1003 2003 3003 3 1299.0
4 1004 2004 3004 1 899.00
5 1005 2005 3005 1 12499.00


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Data.sql
Share
88 CREATE OR REPLACE TABLE RETAIL_DB.RAW.CLEAN_CUSTOMERS AS
89 SELECT
90 CUSTOMER_ID,
91 COALESCE(NULLIF(TRIM(CUSTOMER_NAME), ''), 'UNKNOWN') AS CUSTOMER_NAME,
92 LOWER(TRIM(EMAIL)) AS EMAIL,
93 TRIM(PHONE) AS PHONE,
94 UPPER(TRIM(CITY)) AS CITY
95 FROM RETAIL_DB.RAW.CUSTOMERS;
96 SELECT * FROM RETAIL_DB.RAW.CLEAN_CUSTOMERS;
Results (just now)
Table
Chart
Pivot
20 rows
108ms
CUSTOMER_ID
2001
2020
CUSTOMER
UNKNOWN 10.0%
Aarav Mehta 5.0%
+17 more
EMAIL
aarav.mehta@examp... 5.0%
aditya.bose@exampl... 5.0%
+18 more
PHONE
98765... 5.0%
98765... 5.0%
+18 more
1 2001 Aarav Mehta aarav.mehta@example.com 9876501001
2 2002 Diya Sharma diya.sharma@example.com 9876501002
3 2003 Kabir Rao kabir.rao@example.com 9876501003
4 2004 UNKNOWN customer04@example.com 9876501004
5 2005 Ananya Iyer ananya.iyer@example.com 9876501005


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Share
LOWER(TRIM(EMAIL)) AS EMAIL,
TRIM(PHONE) AS PHONE,
UPPER(TRIM(CITY)) AS CITY
FROM RETAIL_DB.RAW.CUSTOMERS;
SELECT * FROM RETAIL_DB.RAW.CLEAN_CUSTOMERS;
CREATE OR REPLACE TABLE RETAIL_DB.RAW.CLEAN_PRODUCTS AS
SELECT
PRODUCT_ID,
COALESCE(NULLIF(TRIM(PRODUCT_NAME), ''), 'UNKNOWN') AS PRODUCT_NAME,
UPPER(TRIM(CATEGORY)) AS CATEGORY,
UNIT_PRICE,
STOCK_QUANTITY
FROM RETAIL_DB.RAW.PRODUCTS;
SELECT * FROM RETAIL_DB.RAW.CLEAN_PRODUCTS;
Results (just now)
Table
Chart
Pivot
20 rows
111ms
PRODUCT_ID
3001
3020
PRODUCT_NAME
UNKNOWN 10.0%
Bluetooth S... 5.0%
+17 more
CATEGORY
ACCESSO... 35.0%
AUDIO 15.0%
+5 more
UNIT_PRICE
399
18999
STOCK
15
1 3001 Wireless Mouse ACCESSORIES 599.00
2 3002 Mechanical Keyboard ACCESSORIES 2499.00
3 3003 USB-C Hub ACCESSORIES 1299.00
4 3004 UNKNOWN FURNITURE 899.00
5 3005 Monitor ELECTRONICS 12499.00


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My Workspace › Data.sql
ACCOUNTADMIN • RETAIL_WH (X-Small)
RETAIL_DB RAW
Share
SELECT *
FROM RETAIL_DB.RAW.CLEAN_SALES
ORDER BY ORDER_ID;
SELECT
ORDER_ID,
CUSTOMER_ID,
PRODUCT_ID,
QUANTITY,
UNIT_PRICE,
TOTAL_AMOUNT,
CITY AS ORDER_LOCATION
FROM RETAIL_DB.RAW.CLEAN_SALES
ORDER BY ORDER_ID;
Results (just now)
Table
Chart
Pivot
20 rows
51ms
ORDER_ID
CUSTOMER_ID
PRODUCT_ID
QUANTITY
UNIT_PRICE
TOTAL_AMOUNT
ORDER_LOCATION
CHENNAI 25.0%
AHMEDABAD 5.0%
+14 more
1 1001 2001 3001 2 599.00 1198.00 CHENNAI
2 1002 2002 3002 1 2499.00 2499.00 BENGALURU
3 1003 2003 3003 3 1299.00 3897.00 HYDERABAD
4 1004 2004 3004 1 899.00 899.00 CHENNAI
5 1005 2005 3005 1 12499.00 12499.00 COIMBATORE
Feedback


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Data.sql
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My Workspace › Data.sql
ACCOUNTADMIN • RETAIL_WH (X-Small)
RETAIL_DB RAW
Share
SELECT
P.PRODUCT_ID,
P.PRODUCT_NAME,
SUM(S.QUANTITY) AS TOTAL_QUANTITY_SOLD,
SUM(S.TOTAL_AMOUNT) AS TOTAL_SALES
FROM RETAIL_DB.RAW.CLEAN_SALES S
INNER JOIN RETAIL_DB.RAW.CLEAN_PRODUCTS P
ON S.PRODUCT_ID = P.PRODUCT_ID
GROUP BY
P.PRODUCT_ID,
P.PRODUCT_NAME
ORDER BY TOTAL_SALES DESC;
Results (just now)
Table
Chart
Pivot
20 rows
171ms
PRODUCT_ID
PRODUCT_NAME
UNKNOWN 10.0%
Bluetooth Speaker 5.0%
+17 more
TOTAL_QUANTITY_SOLD
TOTAL_SALES
1 3015 Tablet 1 18999.00
2 3005 Monitor 1 12499.00
3 3009 External SSD 1 5999.00
4 3019 Microphone 1 4599.00
5 3008 Bluetooth Speaker 2 4398.00
Feedback

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My Workspace › Data.sql

ACCOUNTADMIN • RETAIL_WH (X-Small) | RETAIL_DB | RAW | Share

```sql
SELECT
    C.CUSTOMER_ID,
    C.CUSTOMER_NAME,
    C.CITY,
    SUM(S.TOTAL_AMOUNT) AS TOTAL_CUSTOMER_SPEND
FROM RETAIL_DB.RAW.CLEAN_SALES S
INNER JOIN RETAIL_DB.RAW.CLEAN_CUSTOMERS C
    ON S.CUSTOMER_ID = C.CUSTOMER_ID
GROUP BY
    C.CUSTOMER_ID,
    C.CUSTOMER_NAME,
    C.CITY
ORDER BY TOTAL_CUSTOMER_SPEND DESC;
```

Results (just now)

Table | Chart | Pivot — 20 rows — 188ms

	CUSTOMER_ID (2001 – 2020)	CUSTOMER_NAME (UNKNOWN 10.0%, Aarav Mehta 5.0%, +17 more)	CITY (CHENNAI 25.0%, AHMEDABAD 5.0%, +14 more)	TOTAL_CUSTOMER_SPEND (899 – 18999)
1	2015	Aditya Bose	KOLKATA	18999.00
2	2005	Ananya Iyer	COIMBATORE	12499.00
3	2009	Tara Menon	CHENNAI	5999.00
4	2019	Dev Malhotra	GURUGRAM	4599.00
5	2008	Ishaan Kapoor	DELHI	4398.00

Feedback

Home | Data.sql | Untitled.sql | Focus

My Workspace › Data.sql

ACCOUNTADMIN • RETAIL_WH (X-Small) | RETAIL_DB | RAW | Share

```sql
ORDER BY TOTAL_CUSTOMER_SPEND DESC;

SELECT
    PRODUCT_ID,
    SUM(TOTAL_AMOUNT) AS TOTAL_SALES
FROM RETAIL_DB.RAW.CLEAN_SALES
GROUP BY PRODUCT_ID
HAVING SUM(TOTAL_AMOUNT) > 5000
ORDER BY TOTAL_SALES DESC;
```

Results (just now)

Table | Chart | Pivot — 3 rows — 278ms

	PRODUCT_ID	TOTAL_SALES
1	3015	18999.00
2	3005	12499.00
3	3009	5999.00

Feedback